National University of Singapore
Department of Statistics and Data Science
Faculty of Science

# Tutorial 8

## Detecting Multicollinearity

ST3131 AY25/26 Sem2

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# 0 What is MC?

**Definition 1** (Multicollinearity)**.** We fit a $k$ regressors MLR $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n\times 1}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}_{p\times 1} \quad \text{and} \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}_{n\times p}.$$

If we find linear combinations in the columns of $\mathbf{X}$, for example

$$\mathbf{X}_1 \approx c_2\mathbf{X}_2 + c_3\mathbf{X}_3 + \cdots + c_p\mathbf{X}_p$$

we call the *multicollinearity* is present is this dataset.

0 What is MC?

1 How to detect MC?

2 Ridge regression as a remedy

Once $\mathbf{X}_1 \approx c_2\mathbf{X}_2 + c_3\mathbf{X}_3 + \cdots + c_p\mathbf{X}_p$ exists:

| **Phenomenon** | **Examination** |
| --- | --- |
| $\exists$ $\boldsymbol{Cor}(\mathbf{X}_j, \mathbf{X}_k)$ very high | $\boldsymbol{Cor}(\mathbf{X}_j, \mathbf{X}_k)$ |
| $\exists$ a good regression $\mathbf{X}_j \sim$ other $\mathbf{X}_{-j}$ | $\mathrm{VIF}_j := \dfrac{1}{1 - R_j^2} > 5$ or $10$ |
| $\mathbf{X}'\mathbf{X}$ has eigenvalue *nearly* 0 | $\dfrac{\max(\text{eigen of } \mathbf{X}'\mathbf{X})}{\min(\text{eigen of } \mathbf{X}'\mathbf{X})} > 100$ or $1k$ *(condition number)* |

**Question 1.** (`Delivery_Time.csv`)

**(a) Find the sample correlation between cases ($X_1$) and distance ($X_2$)**

**(b) Find VIF.**

**(c) Find the condition number of $\mathbf{X'X}$ (in the <u>correlation form</u>).**

## Question 1. (Delivery_Time.csv)

### (a) Find the sample correlation between cases ($X_1$) and distance ($X_2$)

| y | x1 | x2 |
|---|---|---|
| 16.68 | 7 | 560 |
| 11.50 | 3 | 220 |
| 12.03 | 3 | 340 |
| … | … | … |

```r
data = read.csv("Delivery_Time.csv")
data = data[, -1]
colnames(data) = c("y", "x1", "x2")
cor(data$x1, data$x2)
```

```
> cor(data$x1, data$x2)
[1] 0.824215
```

**Question 1. (`Delivery_Time.csv`)**

**(b) Find VIF.**

$$\mathrm{VIF}_1 = \frac{1}{1 - R_1^2} \longleftarrow \texttt{lm(x1\~{}x2)}$$

```r
# Manually compute VIF
reg_x1_x2 = lm(x1~x2, data = data)
Rsq_1 = summary(reg_x1_x2)$r.squared
VIF_1 = 1 / (1-Rsq_1)
```

```
> VIF_1
[1] 3.118474
```

| y | x1 | x2 |
|---|---|---|
| 16.68 | 7 | 560 |
| 11.50 | 3 | 220 |
| 12.03 | 3 | 340 |
| … | … | … |

## Question 1. (Delivery_Time.csv)

## (b) Find VIF.

```r
# Manually compute VIF
reg_x1_x2 = lm(x1~x2, data = data)
Rsq_1 = summary(reg_x1_x2)$r.squared
VIF_1 = 1 / (1-Rsq_1)
```

```r
reg_x2_x1 = lm(x2~x1, data = data)
Rsq_2 = summary(reg_x2_x1)$r.squared
VIF_2 = 1 / (1-Rsq_2)
```

| y | x1 | x2 |
|---|---|---|
| 16.68 | 7 | 560 |
| 11.50 | 3 | 220 |
| 12.03 | 3 | 340 |
| … | … | … |

```
> VIF_1
[1] 3.118474
```

```
> VIF_2
[1] 3.118474
```

$> 5$ or $10$ ?

## Question 1. (Delivery_Time.csv)

## (b) Find VIF.

```
# Manually compute VIF
reg_x1_x2 = lm(x1~x2, data = data)
Rsq_1 = summary(reg_x1_x2)$r.squared
VIF_1 = 1 / (1-Rsq_1)
```

```
# Easy way ^-^
reg = lm(y~x1+x2, data = data)
library(car)
vif(reg)
```

```
reg_x2_x1 = lm(x2~x1, data = data)
Rsq_2 = summary(reg_x2_x1)$r.squared
VIF_2 = 1 / (1-Rsq_2)
```

```
> vif(reg)
      x1       x2
3.118474 3.118474
```

```
> VIF_1
[1] 3.118474
```

```
> VIF_2
[1] 3.118474
```

**Question 1.** (`Delivery_Time.csv`)

**(c) Find the condition number of $\mathbf{X}'\mathbf{X}$ (in the correlation form).**

Set $z_{i1} = \dfrac{x_{i1} - \bar{x}_1}{\sqrt{S_{x1x1}}}$, where $S_{x1x1} = \sum_{i=1}^{n} (x_{i1} - \bar{x}_1)^2$

Set $z_{i2} = \dfrac{x_{i2} - \bar{x}_2}{\sqrt{S_{x2x2}}}$, where $S_{x2x2} = \sum_{i=1}^{n} (x_{i2} - \bar{x}_2)^2$

| z1 | z2 |
| --- | --- |
| -0.05 | 0.09 |
| -0.17 | -0.11 |
| -0.17 | -0.04 |
| … | … |

**Why are we doing so?**

By doing so, the new data matrix $\mathbf{Z}'\mathbf{Z} = samp\mathbf{Cor}(\mathbf{X})$

**Question 1.** (`Delivery_Time.csv`)

**(c) Find the condition number of $\mathbf{X}'\mathbf{X}$ (in the correlation form).**

Set $z_{i1} = \dfrac{x_{i1} - \bar{x}_1}{\sqrt{S_{x1x1}}}$, where $S_{x1x1} = \sum_{i=1}^{n} (x_{i1} - \bar{x}_1)^2$

$$\boxed{\textbf{How to do it quickly?}}$$

```r
z1 = scale(data$x1)/sqrt(24)
```

$$samp\mathbf{Var}(\mathbf{X}_1) \times (n-1)$$

`scale(X₁)`

$$\mathbf{Z}_1 = \frac{\mathbf{X}_1 - \bar{\mathbf{X}}_1}{\sqrt{\text{sampVar}(\mathbf{X}_1) \times (n-1)}} = \frac{\mathbf{X}_1 - \bar{\mathbf{X}}_1}{\sqrt{\text{sampVar}(\mathbf{X}_1)}} \div \sqrt{n-1}$$

**Question 1.** (`Delivery_Time.csv`)

**(c) Find the condition number of $\mathbf{X'X}$ (in the <u>correlation form</u>).**

```r
z1 = scale(data$x1)/sqrt(24)
z2 = scale(data$x2)/sqrt(24)
Z = cbind(z1, z2)
```

By doing so, the new data matrix $\mathbf{Z'Z} = samp\mathbf{Cor}(\mathbf{X})$

| z1 | z2 |
|---|---|
| -0.05 | 0.09 |
| -0.17 | -0.11 |
| -0.17 | -0.04 |
| … | … |

**Question 1.** (`Delivery_Time.csv`)

**(c) Find the condition number of $\mathbf{X'X}$ (in the correlation form).**

```
z1 = scale(data$x1)/sqrt(24)
z2 = scale(data$x2)/sqrt(24)
Z = cbind(z1, z2)
max(eigen(t(Z)%*%Z)$values) /
    min(eigen(t(Z)%*%Z)$values)
```

```
> max(eigen(t(Z) %*% Z)$values) /
   min(eigen(t(Z) %*% Z)$values)
[1] 10.37754
```

$\boxed{\text{Definition of Condition Num}}$

$\boxed{> 100 \text{ or } 1k\ ?}$

**Question 1.** (`Delivery_Time.csv`)

**(a) Find the sample correlation between cases ($X_1$) and distance ($X_2$)**

$\boxed{0.82}$

**(b) Find VIF.**

```
> vif(reg)
      x1         x2
3.118474   3.118474
```

$> 5$ or $10$ ?

**(c) Find the condition number of X′X (in the correlation form).**

```
> max(eigen(t(Z) %*% Z)$values) /
   min(eigen(t(Z) %*% Z)$values)
[1] 10.37754
```

$> 100$ or $1k$ ?

**Question 2.** (Hald_cement_data.csv)

**(a) Fit** $y \sim x_1 + x_2 + x_3 + x_4$ **, get summary and model checking.**

```
data = read.csv("Hald_cement_data.cs
data = data[, -1]
colnames(data) = c("y","x1","x2","x3
reg = lm(y~., data = data)
summary(reg)
```

```
SR = rstandard(reg)
# For QQPlot
qqnorm(SR); qqline(SR)
# For residual plots
plot(x = reg$fitted.values, y = SR)
plot(x = data$x1, y = SR)
......
```

```
Call: lm(formula = y ~., data = data)

Coefficients:
            Estimate   Std. Error   t value   Pr(>|t|)
(Intercept)   62.4     70.0710      0.891     0.3991
x1           1.5511     0.7448      2.083     0.0708 .
x2           0.5102     0.7238      0.705     0.5009
x3           0.1019     0.7547      0.135     0.8959
x4          -0.1441     0.7091     -0.203     0.8441
---
Residual standard error: 2.446 on 8 degrees of freedom
Multiple R-squared: 0.9824, Adjusted R-squared: 0.9736
F-statistic: 111.5 on 4 and 8 DF, p-value: 4.756e-07
```

**Question 2.** (`Hald_cement_data.csv`)

**(b) Compute correlation matrix. Can find MC?**

```
X = data[,c("x1","x2","x3","x4")]
cor(X)
```

```
> cor(X)
        x1     x2     x3     x4
x1   1.000  0.229 -0.824 -0.245
x2   0.229  1.000 -0.139 -0.973
x3  -0.824 -0.139  1.000  0.030
x4  -0.245 -0.973  0.030  1.000
```

**Question 2.** (`Hald_cement_data.csv`)

**(c) Compute VIF. Can find MC?**

$\boxed{> 5 \text{ or } 10\ ?}$

```r
reg = lm(y~., data = data)
summary(reg)
vif(reg)
```

```
> vif(reg)
       x1        x2        x3        x4
 38.49621 254.42317  46.86839 282.51286
```

**Question 2.** (Hald_cement_data.csv)

**(d) Compute condition number (<u>correlation form</u>). Can find MC?**

```
z1 = scale(data$x1) / sqrt(11)
z2 = scale(data$x2) / sqrt(11)
z3 = scale(data$x3) / sqrt(11)
z4 = scale(data$x4) / sqrt(11)
Z = cbind(z1, z2, z3, z4)
eigen(t(Z)%*%Z)$values
max(eigen(t(Z)%*%Z)$values) /
    min(eigen(t(Z)%*%Z)$values)
```

```
> max(eigen(t(X) %*% X)$values) /
   min(eigen(t(X) %*% X)$values)
[1] 1376.881
```

$\boxed{> 100 \text{ or } 1k \text{ ?}}$

0 What is MC?

1 How to detect MC?

2 Ridge regression as a remedy

Dataset $(x_{i1}, \ \dots, x_{ip}, y_i)_{i=1}^n$ $\longrightarrow$ A linear prediction: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$

**Linear regression**

$$\min_{\hat{\beta}'s} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Minimizing $SS_{Res}$

**Ridge regression**

$$\min_{\hat{\beta}'s} \left\{ \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \left( \hat{\beta}_1^2 + \cdots + \hat{\beta}_p^2 \right) \right\}$$

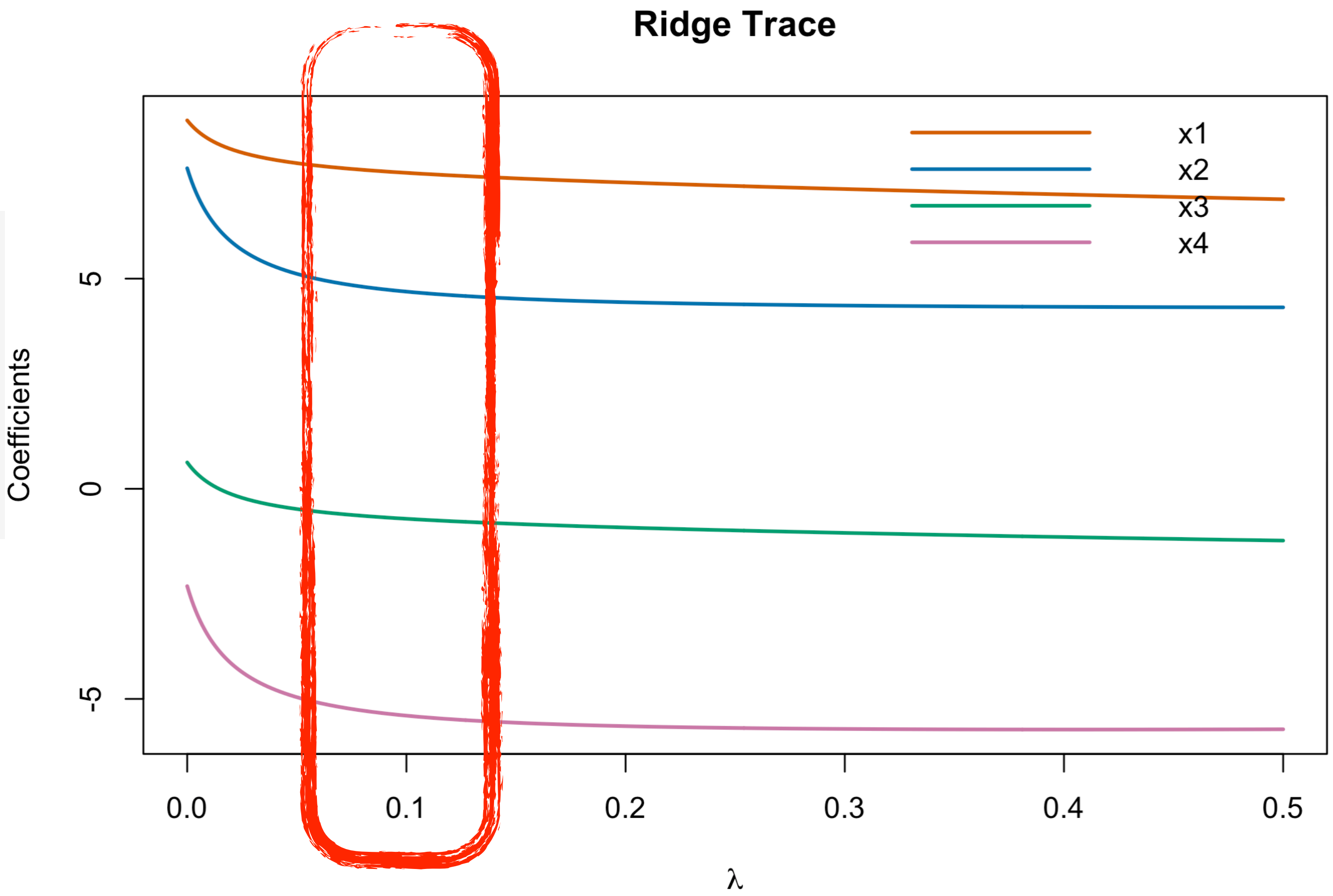
Make $SS_{Res}$ small but also keep $\hat{\beta}'s$ small

$\boxed{\text{Price: choosing } \lambda}$

**Question 2.** (`Hald_cement_data.csv`)

**(e) Use a ridge trace to select $\lambda$. Fit a new model using the chosen value.**

```
grid = seq(0, 0.5, by = 0.001)
library(MASS)
ridge_try = lm.ridge(y~., data = data,
                     lambda = grid)
plot(ridge_try)
```



**Question 2.** (`Hald_cement_data.csv`)

**(e) Use a ridge trace to select $\lambda$. Fit a new model using the chosen value.**

```
# refit by lambda = 0.1
ridge = lm.ridge(y~., data = data,
                 lambda = 0.1)
b = coef(ridge)
# Don't use ridge$coef!!!
```

```
> print(b)
 (Intercept)      x1      x2      x3      x4
      81.825   1.330   0.314  -0.116  -0.336
```

Ridge fitted line: $\hat{y} = 81.825 + 1.33x_1 + 0.314x_2 - 0.116x_3 - 0.336x_4$

**Question 2.** (`Hald_cement_data.csv`)

**(f) Compare $SS_{Res}$ of normal regression and ridge, and $R^2$.**

```
b = coef(ridge)
yhat_ridge = b[1] +
  b["x1"] * data$x1 +
  b["x2"] * data$x2 +
  b["x3"] * data$x3 +
  b["x4"] * data$x4
SSRes_ridge = sum((data$y - yhat_ridge)^2)
```

```
# SSRes for original regression
SSRes_reg = sum(reg$residuals^2)
```

```
> SSRes_ridge
[1] 48.41996
```

```
> SSRes_reg
[1] 47.86364
```

**Question 2.** (`Hald_cement_data.csv`)

**(f) Compare $SS_{Res}$ of normal regression and ridge, and $R^2$.**

Ridge regression

$$\min_{\hat{\beta}'s} \left\{ \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \left( \hat{\beta}_1^2 + \cdots + \hat{\beta}_p^2 \right) \right\}$$

Make $SS_{Res}$ small but also keep $\hat{\beta}'s$ small

```
> SSRes_ridge
[1] 48.41996
```

Linear regression

$$\min_{\hat{\beta}'s} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Minimizing $SS_{Res}$

```
> SSRes_reg
[1] 47.86364
```

## 0 What is MC?

$$\mathbf{X}_1 \approx c_2\mathbf{X}_2 + c_3\mathbf{X}_3 + \cdots + c_p\mathbf{X}_p$$

## 1 How to detect MC?

$$\boldsymbol{Cor}(\mathbf{X}_j, \mathbf{X}_k) \qquad \mathrm{VIF}_j := \frac{1}{1 - R_j^2} \qquad \frac{\max(\text{eigen of } \mathbf{X}'\mathbf{X})}{\min(\text{eigen of } \mathbf{X}'\mathbf{X})}$$

## 2 Ridge regression as a remedy